

THE STRICT LAYER OF CUBICAL TYPE THEORY: CELLULAR REALIZATION, DE MORGAN CONVEXITY, AND THE IMPOSSIBILITY OF NATURAL STRICT FILLERS

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ABSTRACT. In cubical type theories with a De Morgan interval, connections and reversals give types strictly more definitional structure than the identity types of Martin-Löf type theory. Call the layer generated by cell variables under pure reparametrization — path abstraction and application along free De Morgan formulas, with no Kan operations — the *strict layer*. This paper determines it completely and locates its boundary from both sides. (1) Over contexts of generic loops, the strict layer is a wedge of *De Morgan spheres* — boundary-collapsed representables on the De Morgan cube category, acted on strictly functorially. (2) Over an arbitrary cellular context (a cubical computad), it is exactly the *cellular presheaf* of the computad: attachment is definitional, gluing identifies exactly the shared faces, and the spheres are the boundary-degenerate limit; a by-product invisible to the Kan world’s uniqueness up to homotopy is that a boundary can admit several distinct strict fillers. (3) The non-uniqueness resolves fiberwise: the De Morgan convex combination $(\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi')$ — its third term the exact price of the missing excluded middle — makes the algebraically specified correction tower over each algebraic boundary strictly contractible; the algebraic specification is sharp at every level, a loop’s two orientations sharing a boundary yet admitting no strict homotopy, and term-level higher homotopies disconnecting one level up. (4) Yet no coherent choice is possible: *no section of the boundary-fiber correspondence is natural with respect to substitution* — a three-line proof, shown to be widespread by an exhaustive finite census in dimension 2 — so any substitution-stable choice of full-boundary extensions over a generic edge is forced out of the strict layer. In fiber coordinates the convex structure is a graded Boolean median whose iterated cylinders compose on the nose. All displayed terms and concrete conversion equalities and separations cited in the paper are machine-checked in a self-contained proof-assistant kernel; the universally quantified non-existence statements and the general-dimension results are pen-and-paper theorems resting on the explicit kernel interface (K1)–(K3), under which every concrete claim about the calculus is a decidable, assumption-free fact about the algorithmic equality of a minimal fragment.

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1. INTRODUCTION

A basic discovery of homotopy type theory is that the identity types of Martin-Löf type theory equip every type with the structure of a weak ω -groupoid [7, 14]. The structure is *weak*: associativity, units, inverses and the exchange laws hold only up to higher cells, and the bookkeeping of those higher cells — the coherence problem — is notoriously difficult.

Cubical type theories [3] change the balance. Their interval $\mathbb{I}$ carries the operations of a De Morgan algebra ($\wedge, \vee, \neg$), and *reparametrizing* a cell along these operations — forming $\langle i \rangle p(i \wedge j)$ and its relatives, the *connections* — is a definitional operation: many laws that Book HoTT can only prove propositionally hold judgmentally. Practitioners know this well [15]; what has been missing is a precise statement of *how much* of the ω -groupoid structure the definitional equality supplies, where exactly the strict layer ends, and *why* a natural full-boundary extension discipline cannot be confined to it. This paper answers all three questions. Throughout, the *strict layer* over a context means the pure-reparametrization layer: the cubes definable from the context’s cell variables by path abstraction and application along free De Morgan formulas, with the Kan operations excluded. (Terms involving `hcomp` or `transp` also have definitional normal forms; their study belongs to the *weak* layer and to Section 11.)

The wedge of spheres. Fix base types and points, and let q be a generic n -cell — a variable of the n -fold iterated loop type. The definable cubes over such a context, using only reparametrization (no Kan operations), carry an action of the *De Morgan cube category* $\square_{\text{DM}}$: objects $[m]$, morphisms $[m] \rightarrow [n]$ the n -tuples of free De Morgan polynomials in m variables, composition by substitution. This is the cube category with connections, reversals, degeneracies and diagonals, the maximal “classical” cube category in the taxonomy of [5, 2]. Our first main theorem (Theorem 5.9) states: over a context of generic cells $q_1, \dots, q_s$ of arities $n_1, \dots, n_s$, the pure-reparametrization cubes modulo definitional equality form, naturally in $[m]$, the wedge of pointed sets

$$\bigvee_{t=1}^s \square_{\text{DM}}([m], [n_t]) / \partial,$$

where $\square_{\text{DM}}(-, [n]) / \partial$ collapses to the basepoint exactly the morphisms that factor through a face of $[n]$. Each generic cell spans a *De Morgan sphere*, the spheres of distinct cells meet only at the common basepoint, and nothing else is identified. The action realizing the isomorphism is *strictly* functorial: identities and composition hold definitionally. Three consequences delimit the strict/weak boundary exactly: transposition of a 2-cell’s coordinates is not definitional (the precise definitional shadow of the Eckmann–Hilton argument being weak); the square $q(i \wedge \neg i)(j)$, all four of whose edges are degenerate, is not definitionally constant (the non-Booleanness of the free De Morgan algebra is visible in every dimension); and no definitional equation relates cubes built from distinct cells.

Cellular realization. The collapse of the boundary in the sphere theorem is forced by the choice of generic objects: a loop's faces are reflexivities, and reflexivities absorb everything applied to them. The second group of results (Sections 6–7) treats the unpointed situation: a *cellular context*, or *cubical computad* — generic vertices, generic edges between them, generic squares with prescribed edges, and so on — the type-theoretic incarnation of a computad (polygraph) [1, 9] over the cube category. Every computad C has a *cellular presheaf* $|C|$ on $\square_{\text{DM}}$: one representable per cell, glued along the attachment maps — the cubical analogue of the cell complex a CW-presentation describes. The realization theorem (Theorem 7.5) states that the strict layer over the context of C is $|C|$, naturally in the cube dimension: attachment is definitional (each face of a generic cell computes on the nose to its prescribed boundary cell), the gluing identifies exactly the shared faces, and nothing else is identified. For a single cell with generic boundary the layer is the full representable, with no collapse; the spheres are recovered as the boundary-degenerate limit (Proposition 7.11).

As a by-product we isolate a phenomenon that the Kan world's uniqueness up to homotopy renders invisible: *strict fillers are not unique* (Theorem 7.8) — and both the non-uniqueness and the strict higher homotopies that resolve it live entirely in the reparametrization algebra. The elements $i \wedge \neg i$ and $(i \wedge \neg i) \wedge (j \vee \neg j)$ of the free De Morgan algebra are distinct, yet restrict equally to all four faces of the square: two *different* strict squares can fill the *same* boundary, for the entirely algebraic reason that restriction of free De Morgan polynomials to the boundary of a cube is not injective.

De Morgan convexity. The third group of results (Section 8) shows that this non-uniqueness resolves *inside* the strict layer. The *De Morgan convex combination*

$$\chi(k) = (\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi')$$

yields a strict homotopy rel boundary between any two fillers of one cell over a common *algebraic* boundary — a common fiber of the facewise restriction map on formula tuples — and, iterated, makes the *algebraically specified correction tower* — fillers, then corrections with prescribed face formulas, iterated — over a fixed algebraic boundary strictly contractible (Theorem 8.8) — the strict parallel of the uniqueness of Kan fillers up to homotopy, achieved without any Kan operation. The fiberwise qualification is sharp, not cosmetic: over a generic loop, the two orientations $\langle i \rangle p(i)$ and $\langle i \rangle p(\neg i)$ share a type-theoretic boundary yet admit no strict homotopy rel boundary — and in the circle specialization they are even separated propositionally, by winding (Proposition 8.2); one level up, two term-level homotopies with identical constant faces can likewise admit no strict homotopy between them (Proposition 8.9), so the algebraic prescriptions are forced at every level. For cells with faithful attachments the algebraic and type-theoretic boundaries coincide at level 0. The third term is essential: the naive combination $(\varphi \wedge \neg k) \vee (k \wedge \varphi')$ fails *precisely* by the absence of

excluded middle in the free De Morgan algebra, and the failure is machine-checked as a typing error. Fillers originating from different cells admit no direct strict homotopy, but connect by strict deformations onto the shared subcomplex (a machine-checked zig-zag, Theorem 8.12).

The no-go theorem. The final group of results (Sections 9–10) explains why all of this structure is necessary: *no coherent choice of strict fillers exists*. In the free De Morgan algebras the fibers of the boundary map are Boolean intervals 2^D over antichains of diagonal points, and the canonical least filler fails to commute with substitution in a controlled way — the structure theory developed in the companion paper [10]. Theorem 9.1 closes the question of whether the failure can be repaired by a cleverer choice: it cannot. *No section of the boundary-fiber correspondence — no rule assigning to every boundary one of its strict fillers — is natural with respect to substitution*. The proof takes three lines from one uniquely fillable boundary whose transports under a diagonal and under a connective collapse land at the top and the bottom of a single two-element fiber; an exhaustive finite search in dimension 2 shows that the obstruction is widespread rather than exceptional. Consequently (Corollary 9.4) no substitution-stable operation extending full boundaries over a generic edge can be valued in the strict layer: even where strict fillers exist in abundance, no natural choice among them is possible. We are careful about what this does and does not say. Full-boundary extension — a lifting condition against $\partial\Box^n \hookrightarrow \Box^n$ — is a trivial-fibration-style condition, stronger than open-box Kan filling, and we do not claim a no-go for Kan fibrancy itself; whether an algebraic open-box filling structure induces a natural full-boundary section is left open (Section 13). What is proved is that the reparametrization calculus cannot even choose its own fillers naturally: any natural boundary-extension discipline is forced out of the strict layer.

What replaces the impossible choice is *structure*. In fiber coordinates the De Morgan convex combination is computed to be a graded Boolean median (Proposition 10.1); its compositional coherence is witnessed by explicit *iterated* median cylinders whose faces match on the nose (Theorem 10.2); and every compatible sphere of corrections fills. Read informally in the language of algebraic weak factorization systems [4]: the strict layer supports one-step filling data in abundance (any section will do) but, by the no-go theorem, no substitution-stable choice; the strictly contractible median system is the canonical structure that survives, and the no-go theorem says nothing stronger than this coherent-but-unchoosable structure is available. (We keep the awfs language informal throughout; the precise statements are Theorem 9.1, Corollary 9.4, and Theorem 10.2.)

The comparison closed. Section 11 closes the strict-weak comparison over a common boundary. The constant-tube composition equation, which the no-go theorem of the companion paper [11] shows can never be definitional, holds *propositionally and uniformly* via the canonical filler — what cannot

be definitional is exactly what hfill supplies propositionally — while genuinely weak composites open new propositional classes (winding). Within one algebraic boundary fiber, strict cells are unique up to strict homotopy; constant-tube composites are definitionally new but propositionally strict; genuine composites are new outright (Table 1).

Honest positioning. The positive halves of these characterizations — that the De Morgan laws and the attachment equations hold definitionally — are by design of the theory: the interval *is* a free De Morgan algebra and the kernel decides its equality by normal forms [3]. The content of this paper is the *converse* and its structure: nothing else is identified (faithfulness); what collapses is exactly face-factoring; the identifications over a cellular context are exactly those of the cellular presheaf; the interpolant is boundary-constant exactly when restrictions agree; the naive interpolant fails exactly by no-LEM; strict homotopy uniqueness holds despite definitional non-uniqueness; and no natural choice of fillers exists at all. The one-dimensional case of the sphere theorem (a single generic path) appeared as a self-contained appendix of the companion paper [11]; the present paper strictly generalizes it, and beyond that special case we know of no prior statement of any of these results. They do not follow from published normalization theorems: the normalization result of Sterling–Angiuli [12] concerns *Cartesian* cubical type theory, whose cube category has no connections or reversals, and open-term normalization for the De Morgan calculus is not in the literature. Our proofs are elementary given the right lemmas — they analyze the normalization-by-evaluation conversion of a concrete kernel — and every statement about the calculus is, by the metatheory of [11], a decidable and assumption-free fact about a minimal fragment. The cellular results are mechanized in dimension ≤ 2 ; the general-dimension statements are proved by an induction written out in Section 7.1, which consumes no kernel fact beyond (K1) and (K2).

Relation to the companion papers. This paper consolidates and supersedes four preprints of the author (the sphere, realization, convexity, and no-go preprints) distributed with the artifact repository. It is part of a research program whose metatheoretic platform — the kernel, the decidability theorem for the minimal fragment, and the transport-layer no-go theorem — is the companion paper [11], and whose algebraic-combinatorial platform — the fiber structure of free De Morgan and Kleene algebras — is the companion paper [10]. From [11] we import three kernel facts and the decidability theorem (Section 3); from [10] we import the Boolean-interval structure of boundary fibers and the defect calculus (recalled in Section 9). Nothing here depends on the transport-layer design choices of [11] (Remark 3.5).

Outline. Section 2 fixes the De Morgan cube category. Section 3 fixes the calculus, generic cells and cellular contexts, and the imported metatheory. Section 4 proves the strict functoriality of the reparametrization action. Section 5 proves the wedge-of-spheres theorem for generic loops. Section 6 proves that attachment is definitional; Section 7 the realization theorem,

the non-uniqueness of strict fillers, and the sphere recovery. Section 8 develops De Morgan convexity: strict homotopies, contractibility of fiber towers, and the cross-cell zig-zag. Section 9 proves the no-go theorem for natural fillers; Section 10 computes the median correction structure in fiber coordinates. Section 11 closes the strict-weak comparison. Section 12 describes the machine verification.

2. THE DE MORGAN CUBE CATEGORY

Definition 2.1. $\text{DM}(i_1, \dots, i_m)$ is the free De Morgan algebra on generators $i_1, \dots, i_m$: the free bounded distributive lattice with an involution $\neg$ satisfying the De Morgan laws ($\neg(x \wedge y) = \neg x \vee \neg y$, $\neg 0 = 1$), and *without* the laws $x \wedge \neg x = 0$, $x \vee \neg x = 1$. Its elements have canonical normal forms: antichains of clauses over the literals $i_k, \neg i_k$ (disjunctive normal form, reduced under absorption) [3]. We write $0, 1$ for the bounds and call them the *endpoints*.

Definition 2.2 (The De Morgan cube category). $\square_{\text{DM}}$ has objects $[m]$ for $m \geq 0$ and morphisms

$$\square_{\text{DM}}([m], [n]) := \text{DM}(i_1, \dots, i_m)^n,$$

an n -tuple $\vec{\varphi} = (\varphi_1, \dots, \varphi_n)$ regarded as a substitution; composition is substitution of tuples into variables, and the identity on $[n]$ is the tuple of generators $(i_1, \dots, i_n)$.

$\square_{\text{DM}}$ is the classical cube category with faces, degeneracies, symmetries, diagonals, connections of both kinds, and reversals — in the taxonomy of Buchholtz–Morehouse [2] it is the cube category of the “De Morgan” variety; the study of cubical sites with connections goes back to Grandis–Mauri [5]. Two classes of morphisms play a role below:

Definition 2.3. For $\varepsilon \in \{0, 1\}$, the k -th face at ε is the morphism $\delta_k^\varepsilon = (i_1, \dots, i_{k-1}, \varepsilon, i_k, \dots, i_{n-1}) : [n-1] \rightarrow [n]$. A morphism $\vec{\varphi} : [m] \rightarrow [n]$ is *face-factoring* if it factors through some δ_k^ε ; equivalently, iff some coordinate φ_k is an endpoint of the free algebra (i.e. its normal form is the empty or the full antichain).

Definition 2.4 (De Morgan spheres). $\square_{\text{DM}}(-, [n])/\partial$ is the pointed presheaf obtained from the representable $\square_{\text{DM}}(-, [n])$ by identifying all face-factoring morphisms to a single basepoint.

The subtlety that drives several results below is that face-factoring is a normal-form condition, not a pointwise one: $i \wedge \neg i$ takes the value 0 at both endpoints of i , yet it is *not* the endpoint 0 of the free algebra, so $(i \wedge \neg i, j)$ is not face-factoring.

3. THE CALCULUS, CELL CONTEXTS, AND IMPORTED METATHEORY

3.1. The fragment. We work in a CCHM-style cubical type theory [3] with a De Morgan interval; everything below lives in the *minimal fragment*: Π , Σ , path types, and a base type, with transport and homogeneous composition — no universes, no glueing, no higher inductive types. Until Section 11 the terms under study use only path types over a base type: the fragment of *reparametrization*,

$$p \mapsto \langle i_1 \rangle \cdots \langle i_m \rangle p(\varphi_1) \cdots (\varphi_n), \quad \varphi_k \in \text{DM}(i_1, \dots, i_m),$$

where $p(\varphi)$ denotes path application and $\langle i \rangle$ path abstraction. Kan operations (**transp**, **hcomp**) are *excluded* from the layer under study; they are what the weak layer is made of (Section 11).

Definition 3.1 (Generic cells). Over a fixed base type A with a point $a : A$ (A is a primitive type constant of the fragment; no universe is involved), the n -fold loop type is $\text{Loop}^1 := \text{Path } A a a$ and $\text{Loop}^{k+1} := \text{Path } (\text{Loop}^k) \text{ refl}_k \text{ refl}_k$, where refl_k is the k -fold reflexivity tower. A *generic n -cell* is a context variable $q : \text{Loop}^n$. For $\vec{\varphi} \in \square_{\text{DM}}([m], [n])$ we write

$$A_q(\vec{\varphi}) := \langle i_1 \rangle \cdots \langle i_m \rangle q(\varphi_1) \cdots (\varphi_n),$$

an m -cube over the context.

Definition 3.2 (Cellular contexts). A *cellular context of dimension ≤ 2* over the base type A is a context declaring, in order:

- generic vertices $a_1, \dots : A$;
- generic edges $e : \text{Path } A a a'$ between prescribed vertices;
- generic squares $Q : \text{PathP } (\langle i \rangle \text{Path } A (r i) (s i)) p q$ with four prescribed edges p, q, r, s whose endpoints match at the corners.

We spell out dimension ≤ 2 , which is where our machine evidence lives; the recursive definition for all dimensions is Definition 3.3. We call the data a *cubical computad* C ; it is precisely a polygraph [1, 9] over $\square_{\text{DM}}$ presented type-theoretically. A context of generic cells in the sense of Definition 3.1 is the special case in which all boundary cells are reflexivities over a single vertex.

Definition 3.3 (Cube types and computads in all dimensions). Over the base type A , define the n -cube types by recursion. A 0-boundary is empty, and a 0-cube with it is an element of A . An n -boundary β ($n \geq 1$) over a context assigns to each face δ_k^ε of $[n]$ an $(n-1)$ -cube β_k^ε over that context, such that for $k \neq l$ the cubes β_k^ε and $\beta_l^{\varepsilon'}$ have convertible restrictions to their common $(n-2)$ -face. The type of n -cubes with boundary β is the iterated dependent path type

$$\text{Cube}^n(\beta) := \text{PathP } (\langle i \rangle \text{Cube}^{n-1}(\beta \upharpoonright_1 i)) \beta_1^0 \beta_1^1,$$

where $\beta \upharpoonright_1 i$ is the $(n-1)$ -boundary obtained by applying the transverse boundary cubes β_k^ε ($k \geq 2$) at i in their first coordinate; the compatibility

condition on β is exactly what makes this well-typed. A *cellular context* (*cubical computad*) of dimension $\leq N$ declares finitely many cells $c : \text{Cube}^{n_c}(\beta_c)$ with $n_c \leq N$, in an order compatible with dimension, such that each boundary cube of β_c is a pure-reparametrization cube over the subcontext of cells of dimension strictly below n_c (the lower skeleton). Attachment is thus always to strictly lower-dimensional material, as for polygraphs, and the skeletal condition here matches the skeletal recursion of Definition 3.4 exactly. Definition 3.2 is the case $N = 2$. (Once the realization theorem is proved one dimension down, the boundary data of a cell is precisely a presheaf map $\partial\Box^{n_c} \rightarrow |C^{(<n_c)}|$ from the boundary sphere to the cellular presheaf of the preceding skeleton, so the notion coincides with that of a polygraph over $\Box_{\text{DM}}$; the syntactic definition above involves no such circularity.)

Definition 3.4 (Cellular presheaf). Write $\Box^n := \Box_{\text{DM}}(-, [n])$ for the representable and $\partial\Box^n \subseteq \Box^n$ for the subpresheaf of face-factoring morphisms. The *cellular presheaf* $|C|$ of a computad C is defined by recursion on skeleta, as a pushout in the presheaf category $\widehat{\Box_{\text{DM}}}$: $|C^{(\leq r)}|$ is the pushout of

$$|C^{(<r)}| \longleftarrow \coprod_c \partial\Box^{n_c} \longrightarrow \coprod_c \Box^{n_c}$$

over the r -cells c , where the attaching map sends, on the c -summand, a face-factoring morphism $\delta_k^\varepsilon \circ \vec{\chi}$ to the element of $|C^{(<r)}|$ realized by the boundary cube $\beta_{c,k}^\varepsilon$ reparametrized along $\vec{\chi}$; it is well defined on overlaps of faces by the compatibility condition of Definition 3.3. (The attaching map names elements of $|C^{(<r)}|$ through the realization theorem one dimension down; the simultaneous recursion is well-founded because the theorem at stage $< r$ is established before the presheaf at stage r is used — this is the induction of Section 7.1. In dimension ≤ 2 the boundary cubes are single cells and no forward reference is involved.) When every boundary cube is a single cell — as in Definition 3.2 — the pushout reduces to the quotient description

$$|C|([m]) = \left(\coprod_{c \in C} \Box_{\text{DM}}([m], [n_c]) \right) / \sim,$$

with $\sim$ generated by $(c, \delta_k^\varepsilon \circ \vec{\chi}) \sim (\partial_k^\varepsilon c, \vec{\chi})$ for $\partial_k^\varepsilon c$ the prescribed boundary cell. In either form, $|C|$ is the colimit of the representables along the face inclusions specified by the attachments — the cubical cell complex presented by C .

3.2. Algorithmic equality, and what we import. Definitional equality is taken to be the algorithmic equality $\equiv$ of a concrete normalization-by-evaluation kernel: terms are evaluated to weak-head values (path abstractions become closures; applications of variables become *neutral spines*), and conversion compares values type-directedly, opening binders at fresh generic dimensions and comparing interval arguments of neutral path applications by the canonical antichain normal form of Definition 2.1. The kernel is the

one of the companion paper [11], whose artifact this paper’s probes extend; three imported facts are standing hypotheses:

- (K1) *Endpoint collapse is normal-form exact.* Path application of a *value* at an interval argument r collapses to the stored endpoint iff r is an endpoint of the free De Morgan algebra (the kernel decides $r = 0, 1$ by the antichain normal form, not syntactically). For a value of a *dependent* path type $\mathbf{PathP}(\langle i \rangle T) l u$, the stored endpoints are l, u ; for a neutral spine, the stored endpoint *annotations* of its type.
- (K2) *Conversion of neutrals is spine-wise.* Two neutral spines are convertible iff they have the same head variable, the same length, convertible stored endpoint annotations, and, at each path-application stage, interval arguments equal in the free De Morgan algebra.
- (K3) *Decidability without assumptions.* On the minimal fragment, evaluation and conversion terminate on well-typed input and the algorithmic equality is decidable; no base-theory assumption is involved. (This is the endpoint-blind termination theorem of [11].)

(K1) and (K2) are facts about two specific functions of the kernel, verified there; (K3) is a theorem of [11]. Every displayed concrete conversion instance claimed below — well-typedness, equality, or separation of specific terms — is additionally machine-checked (Section 12); by (K3) each such check is a decision, not a test. A remark on self-containedness: (K1) and (K2) as stated are the *complete* interface between this paper and the kernel — no other property of the implementation is used — and each concrete instance below is re-decided by the public artifact independently of the metatheory of [11]; (K3) enters only through the gloss that such checks are decisions rather than tests. The results of this paper are therefore verifiable from its own text together with the archived artifact, without relying on the unpublished status of the companion papers.

Remark 3.5 (Robustness). The reparametrization layer involves no transport, so the results are insensitive to the transport-layer design choices studied in [11] (in particular to its evaluation-time constancy rule): they hold verbatim for a standard CCHM-style conversion. We state them for our kernel because there they are *unconditional* decidable facts by (K3).

4. THE ACTION AND ITS STRICT FUNCTORIALITY

Reparametrization defines, for each generic n -cell q and each m , a function

$$A_q : \square_{\mathbf{DM}}([m], [n]) \longrightarrow \{m\text{-cubes over } \Gamma\}, \quad \vec{\varphi} \mapsto A_q(\vec{\varphi}).$$

Lemma 4.1 (Unit law, definitionally). $A_q(\text{id}_{[n]}) \equiv q$.

Proof. $A_q(\text{id}) = \langle i_1 \rangle \cdots \langle i_n \rangle q(i_1) \cdots (i_n)$, which is $\equiv q$ by n applications of the η -rule for path types (conversion is type-directed: it opens both sides at generic dimensions and compares the resulting applications, which are literally equal spines). Machine probe: `sqEtaD`. $\square$

Lemma 4.2 (Composition law, definitionally). *For $\vec{\varphi} : [m] \rightarrow [n]$ and $\vec{\psi} : [l] \rightarrow [m]$,*

$$(A_q(\vec{\varphi})) \text{ reparametrized along } \vec{\psi} \equiv A_q(\vec{\varphi} \circ \vec{\psi}),$$

where $\vec{\varphi} \circ \vec{\psi}$ is substitution in $\square_{\text{DM}}$.

Proof. Path application of a path abstraction is β -reduction, which the evaluator implements as substitution of the interval argument into the body; iterating, the left-hand side evaluates to the right-hand side's evaluation. Machine probe: `sqCloneCompD` ($((\langle u \rangle \langle v \rangle q(u)(v))(i \vee j)(i \wedge j) \equiv q(i \vee j)(i \wedge j))$). $\square$

Thus A_q is *strictly* functorial — the action of $\square_{\text{DM}}$ on cubes is by definitional equality, with no coherence cells needed. This is the precise sense in which the connection layer of cubical type theory is “strict algebra”: the entire free De Morgan theory acts on cells on the nose. The same computation applies verbatim to the cells of a cellular context; the dependent versions of the unit and absorption laws are re-checked at square types with generic boundary (probes `gbEtaD`, `gbAbsorbD`).

5. GENERIC LOOPS: THE WEDGE OF DE MORGAN SPHERES

Fix a generic n -cell q . The following computation is the engine of everything.

Lemma 5.1 (Value computation). *Open the m binders of $A_q(\vec{\varphi})$ at generic dimensions. The body $q(\varphi_1) \cdots (\varphi_n)$ evaluates as follows.*

- (1) *If some φ_k is an endpoint of the free algebra, the value is the base point a : the first such coordinate collapses the spine — by (K1), either the application of q itself (yielding a reflexivity tower, through which the remaining applications compute), or the application of an already-neutral spine (yielding its stored endpoint annotation, again a tower) — and towers evaluate through the remaining arguments to a .*
- (2) *Otherwise the value is the full neutral spine $q(\varphi_1) \cdots (\varphi_n)$ with all n interval arguments retained.*

In particular the value is a iff $\vec{\varphi}$ is face-factoring (Definition 2.3).

Proof. By (K1), the collapse at each stage happens iff the argument is a free-algebra endpoint; the case analysis is exhaustive. Machine probes: `sq2DegFstD` (collapse at the first coordinate), `sq2DegSndD` (collapse of a neutral spine at the second), `sq2DegCrossD` (all face-factoring squares are equal). $\square$

Theorem 5.2 (C_n : the sphere of a single cell). *For $\vec{\varphi}, \vec{\psi} \in \square_{\text{DM}}([m], [n])$ (with boundaries agreeing so that the comparison is well-typed):*

$$A_q(\vec{\varphi}) \equiv A_q(\vec{\psi}) \iff \begin{cases} \text{both are face-factoring, or} \\ \text{neither is, and } \vec{\varphi} = \vec{\psi} \text{ in } \square_{\text{DM}}([m], [n]). \end{cases}$$

Hence A_q descends to an injection $\square_{\text{DM}}([m], [n])/\partial \hookrightarrow \{m\text{-cubes}\}/\equiv$ of pointed sets, the basepoint being the constant cube.

Proof. Conversion opens the m binders at generic dimensions and compares the bodies of Lemma 5.1.

($\Leftarrow$) If both tuples are face-factoring, both bodies are the same variable a . If neither is and the tuples are equal, the spines agree stage-wise: same head, same stored endpoint annotations (the same towers), and interval arguments equal in the free algebra, which the antichain normal form decides (K2).

($\Rightarrow$) Suppose conversion accepts. If both bodies are a , both tuples are face-factoring by Lemma 5.1. If both are spines, (K2) forces stage-wise equality of the interval arguments in the free algebra, i.e. $\vec{\varphi} = \vec{\psi}$. A variable cannot be convertible with a spine, so the mixed case is impossible. $\square$

Corollary 5.3 (Transposition is not strict). *For a generic 2-cell, $\langle i \rangle \langle j \rangle q(j)(i) \not\equiv \langle i \rangle \langle j \rangle q(i)(j)$.*

Proof. $(j, i) \neq (i, j)$ in $\square_{\text{DM}}([2], [2])$, and neither is face-factoring. Machine probe with a well-typedness control. $\square$

Corollary 5.4 (No excluded middle in dimension 2). *$\langle i \rangle \langle j \rangle q(i \wedge \neg i)(j) \not\equiv \langle i \rangle \langle j \rangle a$, although all four edges of the left-hand square are degenerate.*

Proof. $i \wedge \neg i$ is not an endpoint of the free algebra (its antichain normal form is the singleton clause $\{i, \neg i\}$), so $(i \wedge \neg i, j)$ is not face-factoring; apply Theorem 5.2 against the (constant) face-factoring class. Machine probes `sq2LEMctr1D` and its companion guard. $\square$

Remark 5.5. Corollary 5.4 is the higher-dimensional face of a familiar dimension-1 fact ($\langle i \rangle p(i \wedge \neg i) \not\equiv \text{refl}$): the free De Morgan algebra is not Boolean, and the definitional equality sees this in every dimension — a square all of whose edges are constant need not be definitionally constant. Section 7 will show that this is the germ of a genuine phenomenon: non-uniqueness of strict fillers.

Observation 5.6 (Syntactic exhaustiveness). Over a context with generic cells $q_1 : \text{Loop}^{n_1}, \dots, q_s : \text{Loop}^{n_s}$ (common base point a) and no Kan operations, every definable body of an m -cube evaluates either to a or to an application spine of exactly one q_t : the syntax has no interval case-split, so a single body term cannot mix two cells.

Theorem 5.7 (Wedge theorem). *For cells q_t, q_u and tuples $\vec{\varphi}, \vec{\psi}$:*

$$A_{q_t}(\vec{\varphi}) \equiv A_{q_u}(\vec{\psi}) \iff \begin{cases} \text{both tuples are face-factoring (any } t, u), \text{ or} \\ t = u, \text{ neither is, and } \vec{\varphi} = \vec{\psi}. \end{cases}$$

Proof. Lemma 5.1 applies to each cell separately. Face-factoring bodies are the same variable a regardless of the cell — the shared basepoint. Non-collapsed spines are compared by (K2), which separates distinct head variables ($t \neq u$), distinct spine lengths ($n_t \neq n_u$), and distinct tuples. Machine probes: `sqxDegD` (cross-cell collapse: $\langle i \rangle \langle j \rangle q(0)(j) \equiv \langle i \rangle \langle j \rangle r(i)(1)$),

head separation, and spine-depth separation (a 1-cell square built with the nonzero formula $(i \wedge \neg i) \wedge (j \wedge \neg j)$ — all four edges degenerate, interior nonzero — against a 2-cell square). $\square$

One more lemma extends the characterization from full applications to all cubes of higher loop types (partial applications).

Lemma 5.8 (Partial applications). *An m -cube valued in $\mathbf{Loop}^d$ definable in the layer is a partial application $\langle \vec{i} \rangle q_t(\varphi_1) \cdots (\varphi_{n_t-d})$ (or a tower). Type-directed conversion at $\mathbf{Loop}^d$ opens d further binders at fresh generic dimensions; since a generic dimension is never a free-algebra endpoint, the comparison reduces to Theorem 5.7 for the extended tuples $(\varphi_1, \dots, \varphi_{n_t-d}, l_1, \dots, l_d)$, which are face-factoring iff the original partial tuples are.*

Proof. Conversion at path types is η -directed: it compares applications at a generic dimension. Iterating d times appends d distinct generic dimensions to the spine, none of them an endpoint. Machine probes: the diagonal $[1] \rightarrow [2]$ (`sqDiagCtrlD`) and the diagonal/anti-diagonal separation. $\square$

Theorem 5.9 (A wedge of De Morgan spheres). *Over a context of generic cells $q_1, \dots, q_s$ of arities $n_1, \dots, n_s$ with common base point, the sets of pure-reparametrization m -cubes modulo algorithmic equality assemble, naturally in $[m] \in \square_{\mathbf{DM}}^{\text{op}}$, into the wedge of pointed presheaves*

$$\{\text{cubes}\} / \equiv \cong \bigvee_{t=1}^s \square_{\mathbf{DM}}(-, [n_t]) / \partial.$$

The isomorphism is induced by the actions A_{q_t} ; naturality is Lemmas 4.1 and 4.2 (the actions are strictly functorial); injectivity is Theorems 5.2 and 5.7; surjectivity is Observation 5.6 with Lemma 5.8. All of this is decidable and assumption-free by (K3).

Remark 5.10 (Boundaries). Our generic cells here are loops: their boundaries are reflexivity towers, and it is exactly this degeneracy that produces the $/\partial$ collapse (Lemma 5.1 computes the collapse *through* the towers). The next two sections treat the unpointed situation — generic cells with *generic* boundaries — and show that the collapse disappears: the layer becomes the full representable, glued exactly along the attachments, with the spheres as the boundary-degenerate limit.

6. ATTACHMENT IS DEFINITIONAL

Throughout this section and the next, fix a cellular context; for a generic square Q with edges p, q, r, s (as in Definition 3.2) and $\varphi, \psi \in \mathbf{DM}(i, j)$ write $Q(\varphi)(\psi)$ for the doubly applied cell (with the boundary-typed endpoint annotations) and $A_Q(\vec{\varphi}) := \langle \vec{i} \rangle Q(\varphi_1)(\varphi_2)$ for the reparametrized cube, and likewise for edges.

Theorem 6.1 (Attachment equations). *For a generic square Q with prescribed edges p, q, r, s :*

$$\langle j \rangle Q(0)(j) \equiv p, \quad \langle j \rangle Q(1)(j) \equiv q, \quad \langle i \rangle Q(i)(0) \equiv r, \quad \langle i \rangle Q(i)(1) \equiv s,$$

and corners compute through two steps: $\langle i \rangle \langle j \rangle Q(0)(0) \equiv \langle i \rangle \langle j \rangle a_{00}$, etc. More generally the whole face square is the edge square: $\langle i \rangle \langle j \rangle Q(0)(j) \equiv \langle i \rangle \langle j \rangle p(j)$.

Proof. By (K1) at the outer application: $Q(0)$ collapses to the stored endpoint of the **PathP** type, which is the *prescribed edge* p — a generic cell, not a reflexivity; the remaining application and η give the first equation. For the third, $Q(\varphi)$ at non-endpoint φ is a neutral spine whose stored endpoint annotations are $r(\varphi), s(\varphi)$; (K1) at the inner application returns them, and η finishes. Corners iterate the two collapses. Machine probes: **gbAttPD**, **gbAttQD**, **gbAttRD**, **gbAttSD**, **gbCornerD**, **gbEdgeSqD**. $\square$

Remark 6.2. This is the exact point at which the generic-boundary situation departs from the loops of Section 5: there the stored endpoints were reflexivity towers, through which every further application computed to the base point — producing the $/\partial$ collapse. Here the stored endpoints are generic cells, and the face-factoring morphisms land in the boundary's own layer instead of a single point. The De Morgan laws themselves remain strict in the presence of a generic boundary: the unit and absorption laws hold at the full dependent square type (machine probes **gbEtaD**, **gbAbsorbD**, **gbAbsorbPerturbD**).

7. THE REALIZATION THEOREM

Lemma 7.1 (Value computation, generic boundary). *Over the square computed (four vertices, four edges, one square), open the m binders of a definable cube at generic dimensions. The body evaluates to exactly one of:*

- (1) a vertex variable;
- (2) an edge spine $e(\chi)$ with $e \in \{p, q, r, s\}$ and χ a non-endpoint;
- (3) a full spine $Q(\varphi)(\psi)$ with both coordinates non-endpoints.

A tuple (φ, ψ) face-factoring as $\delta^\varepsilon \circ \chi$ evaluates to the same value as the corresponding boundary composite: the edge spine of the prescribed face at χ (or a vertex, when χ is again an endpoint).

Proof. (K1) case analysis as in Theorem 6.1; the three cases are exhaustive because collapse consumes coordinates until either none remain (vertex) or the first non-endpoint argument freezes the spine. $\square$

Theorem 7.2 (Single cell, full representable). *The pure-reparametrization m -cubes over the square computed, modulo $\equiv$, are naturally isomorphic to $|C| = \square_{\text{DM}}(-, [2])$ — the full representable, the isomorphism sending $\vec{\varphi}$ to $A_Q(\vec{\varphi})$ and the face-factoring morphisms to the boundary composites.*

Proof. Surjectivity is Lemma 7.1: every value is a vertex, edge, or full spine, i.e. the image of a morphism (for the square computad, every boundary composite is a face-factoring morphism of $[2]$, so $|C|([m]) = \square_{\text{DM}}([m], [2])$ — attachments here *create* no quotient, they only relate the coproduct's summands, and every summand embeds by face composition).

Well-definedness (the presheaf identifications hold) is Theorem 6.1.

Injectivity: two full spines are convertible iff their tuples agree (K2, interval arguments componentwise in the free algebra); two edge spines iff same edge and same argument (K2: the four edges are distinct head variables); vertex values are the four distinct vertex variables; and the three classes are mutually non-convertible (distinct heads; a variable is not a spine). A face-factoring tuple and a non-face-factoring tuple land in different classes by Lemma 7.1. Finally two face-factoring tuples with the same value have the same boundary composite, hence are equal in $\square_{\text{DM}}([m], [2])$ — e.g. $p(\chi) = p(\chi')$ forces $\chi = \chi'$, and cross-edge coincidences occur only at corners, where the tuples agree. $\square$

Remark 7.3 (Higher dimensions). For a generic n -cell attached along the full cube-boundary computad of dimension n , Lemma 7.1 and its consequences go through by induction on n : (K1) peels endpoint coordinates one at a time, each collapse landing in the boundary computad one dimension down, and (K2) separates the resulting spines by head and length. The general statement, for arbitrary computads in all dimensions, is Theorem 7.6, proved in Section 7.1.

The induction step for several cells is the two-square computad: squares Q_1 (edges p, m, r_1, s_1) and Q_2 (edges m, q, r_2, s_2) *sharing* the edge m .

Theorem 7.4 (Gluing along a shared face). (1) (*Identification.*) *The two faces agree across the two cells, definitionally:*

$$\langle i \rangle \langle j \rangle Q_1(1)(j) \equiv \langle i \rangle \langle j \rangle Q_2(0)(j) \quad (\text{both} \equiv \text{the edge square of } m).$$

(2) (*No over-identification.*) *Let $E_0 := i \vee \neg i \vee j \vee \neg j$ and $D_0 := i \wedge \neg i \wedge j \wedge \neg j$ — the edge-one and edge-zero elements of $\text{DM}(i, j)$. The squares*

$$S_1 := \langle i \rangle \langle j \rangle Q_1(E_0)(j), \quad S_2 := \langle i \rangle \langle j \rangle Q_2(D_0)(j)$$

have their entire boundaries on m : together with the edge square $\langle i \rangle \langle j \rangle m(j)$, both inhabit the type $\text{Path}(\text{Path } A a_{10} a_{11}) m m$. All three are pairwise non-convertible.

Proof. (1) Both sides collapse by (K1) to the shared cell m (Theorem 6.1 applied in each square), hence to each other. Machine probes `seAttM1D`, `seAttM2D`, `seGlueD`. (2) E_0 and D_0 are non-endpoints of the free algebra whose restrictions to all four faces are 1 resp. 0; so S_1, S_2 are full spines with heads Q_1 resp. Q_2 , the edge square is an m -spine, and (K2) separates the three by head. Well-typedness at the common type is the boundary

computation of (1). Machine probes `seS1CtrlD`, `seS2CtrlD`, `seSmCtrlD` and the three separation guards. $\square$

Theorem 7.5 (Realization in dimension ≤ 2). *For every cubical computad C of dimension ≤ 2 , the pure-reparametrization m -cubes over the context of C , modulo the algorithmic equality, form naturally in $[m]$ the cellular presheaf:*

$$\{m\text{-cubes}\}/\equiv \cong |C|([m]).$$

All of this is decidable and assumption-free by (K3).

Proof. Induction on the cells of C . Adding a vertex or an edge is the base analysis of Lemma 7.1 (an edge over its two vertices is the one-dimensional instance of Theorem 7.2). Adding a square Q attached along prescribed edges: the new definable bodies are exactly the Q -spines (Lemma 7.1, whose proof is local to the cell and its boundary); the attachment identifications hold definitionally (Theorem 6.1), matching the presheaf's $\sim$; and no new identifications occur — between a Q -spine and anything previously definable (K2, head separation; the same-boundary cases are exactly those of Theorem 7.4(2), which is the general no-merge computation: the boundary-collapsed interior of a new cell never merges with old material or with another cell's interior). Naturality in $[m]$ — compatibility with reparametrization — is the strict functoriality of the actions (Lemmas 4.1 and 4.2, rechecked at dependent square types: `gbEtaD`, `gbAbsorbD`). $\square$

7.1. The general induction.

Theorem 7.6 (Realization, all dimensions). *For every cubical computad C of finite dimension (Definition 3.3), the pure-reparametrization m -cubes over the context of C , modulo the algorithmic equality, form naturally in $[m]$ the cellular presheaf: $\{m\text{-cubes}\}/\equiv \cong |C|([m])$.*

Proof. By induction on a construction order of the cells; the induction hypothesis is the theorem for the preceding context. By Definition 3.3 the boundary of each new cell lies in the lower skeleton, which the preceding context contains, so the hypothesis always covers the attachment target. Cells of dimension ≤ 2 are Theorems 7.2–7.5. Let c be a declared n -cell with boundary β_c .

(i) *Value computation.* A definable body over the extended context is an iterated path application of a single declared variable to interval formulas (the syntax has no interval case-split, so Observation 5.6 applies verbatim). Consider a body $c(\varphi_1) \cdots (\varphi_j)$, $j \leq n$. If every applied coordinate is a non-endpoint, the value is the frozen neutral spine with head c . Otherwise let φ_l be the first endpoint coordinate, say $\varphi_l = \varepsilon$. By (K1) the application at stage l collapses to the stored endpoint of the instantiated telescope, which by Definition 3.3 is the prescribed boundary cube $\beta_{c,l}^\varepsilon$ applied at $\varphi_1, \dots, \varphi_{l-1}$; the remaining applications compute inside this lower-skeleton material, where the induction hypothesis controls the value. The result is

the value of the boundary composite $(\partial_l^\varepsilon c, (\varphi_1, \dots, \widehat{\varphi_l}, \dots, \varphi_j))$ — exactly the generating relation of $|C|$ (Definition 3.4).

(ii) *Consistency across factorizations.* If $\vec{\varphi}$ factors through two different faces, evaluation — being a function — assigns the body a single value, so the two boundary composites have equal values. On the presheaf side the two identifications agree as well: the boundary data of c is a compatible family, so $\beta_{c,k}^\varepsilon$ and $\beta_{c,l}^{\varepsilon'}$ restrict to convertible cubes on their common $(n-2)$ -face (Definition 3.3), and the induction hypothesis identifies equal lower-skeleton values with equal elements of $|C^{(<n)}|$. The same reasoning covers a cell attached to one and the same lower cell along several faces: each face's stored endpoint is whatever β_c prescribes there, and no further compatibility is needed.

(iii) *Surjectivity and well-definedness.* By (i) every value is either lower-skeleton material — by the induction hypothesis in the image of $|C^{(<n)}|$ — or a spine of c , i.e. the image of a morphism of $\square_{\text{DM}}(-, [n])$; partial spines are handled by the η -argument of Lemma 5.8, which appends fresh generic dimensions and is typed at arbitrary telescopes. The presheaf relations hold definitionally: computation (i) is the general attachment equation, obtained from (K1) exactly as in Theorem 6.1.

(iv) *No over-identification.* Two frozen spines with head c are convertible iff their tuples are equal in the free algebra (K2). A frozen spine with head c is never convertible with lower-skeleton material: by the induction hypothesis such material consists of variables and frozen spines of previously declared cells, whose heads differ from c , and (K2) separates distinct heads (and a variable is not a spine). This is the general form of Theorem 7.4(2), and it covers in particular every same-boundary configuration. Finally, two face-factoring tuples with equal values have equal boundary composites by (i) together with the injectivity half of the induction hypothesis applied in the lower skeleton, so face-factoring material is identified in $|C|$ exactly when the values agree.

Naturality in $[m]$ is the strict functoriality of the actions (Lemmas 4.1 and 4.2), whose proofs are typed at arbitrary telescopes. $\square$

Remark 7.7 (Machine evidence). The mechanized probes cover dimension ≤ 2 (Section 12). The induction above consumes, at each stage, only (K1) at one telescope stage, the head and argument discipline of (K2), and the induction hypothesis — no kernel fact beyond those already exercised in the mechanized dimensions. Readers preferring a fully mechanized statement may read every cellular theorem of this paper at dimension ≤ 2 ; no later result uses more.

7.2. Strict fillers are not unique. In the Kan world, a boundary determines its fillers up to homotopy. In the strict layer the situation is sharper in one way — fillers are literal — and looser in another:

Theorem 7.8 (Non-uniqueness of strict fillers). *Over the square computed there exist two non-convertible squares with the same boundary: with $\varphi_1 :=$*

$i \wedge \neg i$ and $\varphi_2 := (i \wedge \neg i) \wedge (j \vee \neg j)$,

$$\langle i \rangle \langle j \rangle Q(\varphi_1)(j) \not\equiv \langle i \rangle \langle j \rangle Q(\varphi_2)(j),$$

yet both inhabit $\text{PathP}(\langle i \rangle \text{Path } A(r(i \wedge \neg i))(s(i \wedge \neg i))) pp$.

Proof. $\varphi_1 \neq \varphi_2$ in the free algebra: absorption fails because the clause $\{i, \neg i\}$ contains neither $\{j\}$ nor $\{\neg j\}$, so $(i \wedge \neg i) \wedge (j \vee \neg j) \neq i \wedge \neg i$. Yet their restrictions to the four faces coincide (0 on the i -faces, $i \wedge \neg i$ on the j -faces), so the two squares have convertible boundaries; being non-degenerate full spines with distinct interval arguments, (K2) separates them. Machine probes `gbFill1D`, `gbFill2D` and the separation guard. $\square$

Proposition 7.9 (The minimal perturbation is absorbed). *The edge-vanishing elements of $\text{DM}(i, j)$ are exactly 0 and $D_0 = i \wedge \neg i \wedge j \wedge \neg j$, and $D_0 \leq \varphi$ for every $\varphi \neq 0$ (the clause of D_0 contains every literal, hence contains every clause). Consequently join-perturbation by D_0 never produces a new filler: $\varphi \vee D_0 = \varphi$ (machine probe `gbAbsorbPerturbD`). The non-uniqueness of Theorem 7.8 therefore arises only through meet-constructions such as φ_2 , i.e. through the non-injectivity of the boundary-restriction map $\text{DM}(i, j) \rightarrow \text{DM}(j)^2 \times \text{DM}(i)^2$ — an intrinsic feature of free De Morgan algebras, faithfully reflected by definitional equality.*

Remark 7.10. Theorem 7.8 is consistent with the realization theorem — the representable $\square_{\text{DM}}(-, [2])$ genuinely has several elements over a single boundary configuration — but it is worth isolating because it contradicts a natural first guess (that strict fillers, when they exist, are unique) and because it gives the boundary-restriction non-injectivity of free De Morgan algebras a geometric meaning inside type theory. Section 8 shows that the non-uniqueness is coherently contractible within the corresponding algebraic correction tower; Section 9 shows that it is nevertheless impossible to choose fillers naturally.

7.3. Recovering the spheres. The generic loops of Section 5 are the boundary-degenerate limit of the present setting: substituting the constant computad (one vertex, reflexivity boundaries) for the boundary cells sends the square computad to the 2-loop context. Under this substitution the stored endpoints of the spine collapse to reflexivity towers, through which (K1) computes to the base point — precisely the mechanism that turns the full representable into the boundary-collapsed one:

Proposition 7.11. *Boundary degeneration realizes the quotient: the substitution of the point computad for the boundary carries the isomorphism of Theorem 7.2 to the sphere isomorphism $\{\text{cubes}\}/\equiv \cong \square_{\text{DM}}(-, [2])/\partial$ of Theorem 5.9, the induced map on presheaves being the canonical collapse $\square_{\text{DM}}(-, [2]) \twoheadrightarrow \square_{\text{DM}}(-, [2])/\partial$.*

Proof. The substitution sends attachment values (edge spines, vertices) to reflexivity towers and the base point; Lemma 7.1 degenerates clause-wise to

the value computation of Lemma 5.1, identifying exactly the face-factoring classes. $\square$

Together, Theorems 7.2 and 7.5 and Proposition 7.11 give a uniform picture: *the definitional equality of cubical type theory realizes cubical computads as their cell complexes, on the nose; collapsing boundaries collapses realizations accordingly.*

8. DE MORGAN CONVEXITY

The realization theorem classifies strict cubes injectively by cells and formula tuples. For a cell whose attachment is *faithful* — distinct faces attached to distinct generic cells, as in the square computad — equality of type-theoretic boundaries forces equality of the tuples' face restrictions, by the injectivity half of the realization theorem applied to the faces. For *self-identifying* attachments this fails — a loop attaches both endpoints to one vertex, and the formulas 0 and 1 restrict to the same boundary *value* while being distinct in the free algebra — and the strict homotopy theory must be organized by the finer, algebraic notion of boundary (Definition 8.1); Proposition 8.2 shows the finer notion is not a technical convenience but sharp. We now develop the strict layer's internal homotopy theory. The running example is the non-uniqueness pair of Theorem 7.8:

$$F_1 := \langle i \rangle \langle j \rangle Q(i \wedge \neg i)(j), \quad F_2 := \langle i \rangle \langle j \rangle Q((i \wedge \neg i) \wedge (j \vee \neg j))(j),$$

whose formula tuples restrict identically to all four faces (0 on the i -faces, $i \wedge \neg i$ on the j -faces): the pair lies over a single algebraic boundary in the sense of the following definition.

8.1. Algebraic boundaries, and interpolation.

Definition 8.1 (Algebraic boundary). Fix a cell c of arity n . An *algebraic boundary* for c in dimension m is a fiber of the facewise-restriction map on formula tuples: two strict fillers $A_c(\vec{\varphi}), A_c(\vec{\varphi}')$ lie *over the same algebraic boundary* if $\vec{\varphi}|_f = \vec{\varphi}'|_f$ in the free De Morgan algebra for every face f of the m -cube. Fillers over one algebraic boundary have convertible type-theoretic boundaries; the converse holds when the attachment of c is faithful (the paragraph opening this section) but fails for self-identifying attachments.

Proposition 8.2 (Sharpness: orientation classes). *Over a generic loop $p : \text{Path } A a a$, the paths $F = \langle i \rangle p(i)$ and $F' = \langle i \rangle p(\neg i)$ have the same type-theoretic boundary (a, a) but lie over distinct algebraic boundaries $((0, 1)$ and $(1, 0))$, and no strict homotopy rel boundary connects them. Indeed, a candidate is a strict square $\langle k \rangle \langle i \rangle p(\psi)$ with $\psi \in \text{DM}(i, k)$ (over the loop context every non-constant square is p -material, by Theorem 5.9), and rel boundary forces $\psi(i, 0) = i$, $\psi(i, 1) = \neg i$, and $\psi(0, k)$ equal to a constant endpoint of $\text{DM}(k)$; but $\psi(0, 0) = 0 \neq 1 = \psi(0, 1)$, a contradiction. The interpolant of Lemma 8.3 indeed fails rel boundary here: $\chi = (i \wedge \neg k) \vee (k \wedge \neg i) \vee (i \wedge \neg i)$ has $i=0$ face $\langle k \rangle p(k)$. The impossibility above is definitional*

and holds over any base type. In the circle specialization ($A = S^1$, $p = \text{loop}$) the two orientations are moreover separated propositionally — winding $+1$ and -1 (Proposition 11.3) — so there not even a weak homotopy connects them rel boundary: the fiberwise organization below is forced, not chosen.

Lemma 8.3 (Interpolation). *For $\varphi, \varphi' \in \text{DM}(\vec{i})$ let*

$$\chi(k) := (\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi') \in \text{DM}(\vec{i}, k).$$

Then:

- (1) $\chi(0/k) = \varphi$ and $\chi(1/k) = \varphi'$;
- (2) for any face f of the $\vec{i}$ -cube on which $\varphi|_f = \varphi'|_f = \rho$: $\chi|_f = \rho$, constant in k .

Proof. (1) $\chi(0) = (\varphi \wedge 1) \vee 0 \vee (\varphi \wedge \varphi') = \varphi \vee (\varphi \wedge \varphi') = \varphi$ by absorption ($\varphi \wedge \varphi' \leq \varphi$); symmetrically at 1. (2) $\chi|_f = (\rho \wedge \neg k) \vee (k \wedge \rho) \vee \rho = \rho$, again by absorption (the first two joinands are $\leq \rho$). $\square$

Theorem 8.4 (Strict homotopy rel boundary). *Let $F = A_c(\vec{\varphi})$ and $F' = A_c(\vec{\varphi}')$ be strict fillers over the same algebraic boundary (Definition 8.1) arising from the same cell c . Then*

$$H := A_c(\vec{\chi}), \quad \chi_t := (\varphi_t \wedge \neg k) \vee (k \wedge \varphi'_t) \vee (\varphi_t \wedge \varphi'_t),$$

with k as the outer dimension, is a strict homotopy rel boundary from F to F' .

Proof. The hypothesis is exactly that $\vec{\varphi}|_f = \vec{\varphi}'|_f$ on every face f ; when the attachment of c is faithful, this follows from equality of the type-theoretic boundaries, so no separate hypothesis is needed there. Lemma 8.3(1) gives the k -endpoints; Lemma 8.3(2), applied coordinatewise, makes every face of H constant in k and equal to the given boundary, so H inhabits the path type over the boundary type from F to F' . (Typing uses only the boundary conditions; interior degeneracies of $\vec{\chi}$ are irrelevant.) Machine witness on the running example: `mixInterpD`, with a control probe and with the separation $F_1 \not\equiv F_2$ re-checked — the homotopy is not trivial. $\square$

Lemma 8.5 (Necessity of the third term). *For k -free $\rho \neq 0$:*

$$(\rho \wedge \neg k) \vee (k \wedge \rho) = \rho \wedge (k \vee \neg k) \neq \rho,$$

since $\rho \leq k \vee \neg k$ would require every clause of ρ to contain k or $\neg k$. Consequently the naive interpolant $(\varphi \wedge \neg k) \vee (k \wedge \varphi')$ is not boundary-constant, and the corresponding term fails to typecheck as a homotopy (machine-checked failure on the running example).

Remark 8.6. Lemma 8.5 is a small but pointed fact: in a Boolean interval the naive convex combination would do, and the third term is invisible. The free De Morgan algebra’s missing excluded middle — the very feature that keeps the connection layer faithful (Theorem 5.2) — exacts the correction term $\varphi \wedge \varphi'$ as its price. The corrected operation is the “strong”

(Kleene-style) convex combination, and Section 10 shows it is exactly a graded Boolean median in fiber coordinates.

8.2. Strict contractibility of fiber towers. Fix a cellular context, a cell c , and an *algebraic* boundary β (Definition 8.1), with face prescriptions β_f .

Definition 8.7 (Algebraic correction tower). The tower is defined at the level of formula tuples. $\text{Fib}^0(\beta)$ is the set of tuples $\vec{\varphi}$ with $\vec{\varphi}|_f = \beta_f$ for every face f ; each is realized by the strict filler $A_c(\vec{\varphi})$. For $x, y \in \text{Fib}^n$, $\text{Fib}^{n+1}(x, y)$ is the set of tuples $\vec{\psi}$ in one further variable k satisfying, *as formulas*: $\vec{\psi}(0/k) = x$, $\vec{\psi}(1/k) = y$, and on every other face f , $\vec{\psi}|_f$ equals the constant (cylinder) extension of the prescribed lower formula; each is realized by a strict homotopy rel boundary between the realizations of x and y . Here Fib^n is a globular family indexed by its complete prescribed algebraic boundary — the pair x, y one level down together with the cylinder prescriptions on all remaining faces; the notation suppresses this indexing. The tower is *strictly contractible* if at every level any two elements over the same prescribed boundary are connected: $\text{Fib}^{n+1}(x, y) \neq \emptyset$ for all x, y . The term is bespoke to this tower: strict contractibility is a levelwise connect-edness property of the concrete sets Fib^n , proved without any ambient Kan or model structure, and it neither presupposes nor asserts the contractibility of a Kan complex or of a categorical object. The formula-level (rather than term-level) prescription of the faces is essential, not cosmetic: Proposition 8.9 exhibits two term-level strict homotopies with identical constant faces that admit no strict homotopy between them — the mechanism of Proposition 8.2, one level up.

Theorem 8.8 (Contractibility). *The algebraic correction tower of a cell over any algebraic boundary is strictly contractible.*

Proof. Level 1: two elements $x, y \in \text{Fib}^0(\beta)$ have $x|_f = y|_f = \beta_f$ on every face, so the interpolant $\vec{\chi}$ of Theorem 8.4 connects their realizations; moreover $\vec{\chi}|_f = \beta_f$ *as a formula* by Lemma 8.3(2) — absorption computes the interpolant of two equal restrictions to exactly that restriction — so $\vec{\chi} \in \text{Fib}^1(x, y)$. For level $n + 1$: by Definition 8.7, any two elements $H, H' \in \text{Fib}^{n+1}(x, y)$ have *identical* formula restrictions on every face — the k -endpoints are the specific tuples x, y and every other face is the prescribed cylinder formula — so the same interpolation applies verbatim one dimension up, and its faces are again the prescribed formulas by the same absorption computation. Induction. $\square$

Proposition 8.9 (Term-level towers can disconnect). *Over a generic 2-loop q , let $d_i = i \wedge \neg i$, $d_j = j \wedge \neg j$, $t = k \wedge \neg k$, and let $X = \langle i \rangle \langle j \rangle q(0)(0)$ be the constant square. The 3-cubes*

$$H = \langle k \rangle \langle i \rangle \langle j \rangle q(t \wedge d_i)(t \wedge d_j), \quad H' = \langle k \rangle \langle i \rangle \langle j \rangle q(t \wedge d_j)(t \wedge d_i)$$

are strict homotopies $X \Rightarrow X$ every one of whose faces is constant — each face substitution sends one coordinate to the endpoint 0 — and $H \not\equiv H'$

by (K2) (distinct non-face-factoring tuples). Yet no strict homotopy $L : H \Rightarrow H'$ rel boundary exists: its tuple $\vec{\psi} \in \text{DM}(l, k, i, j)^2$ would satisfy $\vec{\psi}(0/l) = (t \wedge d_i, t \wedge d_j)$ and $\vec{\psi}(1/l) = (t \wedge d_j, t \wedge d_i)$, so $\psi_1|_{i=0}$ restricts to 0 at $l = 0$ and to $t \wedge d_j$ at $l = 1$, and $\psi_2|_{i=0}$ to $t \wedge d_j$ and 0: neither coordinate of $\vec{\psi}|_{i=0}$ is a global endpoint of the free algebra, so by Lemma 5.1 the $i=0$ face of L is a non-collapsed spine, not the required constant. The mechanism is that of Proposition 8.2 one level up: H and H' have equal face values but unequal face restrictions — $(0, t \wedge d_j)$ versus $(t \wedge d_j, 0)$ at $i = 0$ — i.e. they lie over different algebraic boundaries, and Definition 8.7 rightly excludes such pairs from a common tower.

Remark 8.10. Together with Theorem 7.8, the picture is: *strict fillers are not unique, but, within each algebraic boundary fiber, are unique up to strict homotopy, and the canonical correction system iterates coherently at all levels* — in the algebraically specified sense of Definition 8.7 — the strict parallel of Kan filling’s uniqueness up to homotopy, achieved without any Kan operation. The algebraic specification is forced at every level: across fibers the separation is genuine at level 0 (Proposition 8.2) and at level 1 (Proposition 8.9), and it is exactly this fiber structure that Sections 9–10 study algebraically. We call the underlying canonical structure — the interpolation operator as a convex combination on fibers — *De Morgan convexity*.

8.3. Cross-cell fillers and the shared subcomplex. Fillers of one boundary can originate from different cells: over the shared-edge computad of Theorem 7.4, the squares S_1, S_2 and the edge square $S_m := \langle i \rangle \langle j \rangle m(j)$ all inhabit $\text{Path}(\text{Path } A a_{10} a_{11}) m m$ and are pairwise non-convertible.

Proposition 8.11 (No direct homotopy). *There is no strict homotopy with k -faces S_1 and S_2 : by the realization theorem a strict cube is a single cell’s material, and its two k -faces are that cell’s material or boundary material — never the interiors of two different cells.*

Theorem 8.12 (Connection through the shared subcomplex). *Both interiors deform strictly onto the shared material:*

$$\langle k \rangle \langle i \rangle \langle j \rangle Q_1((i \vee \neg i \vee j \vee \neg j) \vee k)(j) : S_1 \rightsquigarrow S_m,$$

$$\langle k \rangle \langle i \rangle \langle j \rangle Q_2((i \wedge \neg i \wedge j \wedge \neg j) \wedge \neg k)(j) : S_2 \rightsquigarrow S_m,$$

— at $k = 1$ the first formula becomes 1 and $Q_1(1)(j) \equiv m(j)$ by the attachment equations (Theorem 6.1); dually for the second. Hence the strict fiber is connected by the zig-zag $S_1 \rightsquigarrow S_m, S_2 \rightsquigarrow S_m$ — entirely inside the strict layer. (Machine witnesses `mixDeformQ1D`, `mixDeformQ2D`.)

Remark 8.13 (The general pattern). When same-boundary fillers originate from different cells, the realization theorem forces the boundary to lie in the *shared subcomplex*, and the coordinate of each filler transverse to its shared face then has endpoint restrictions on every face — so Lemma 8.3 applies

against the corresponding constant, deforming each filler onto shared material; provided the resulting lower-dimensional fillers carry the same algebraic face prescriptions, Theorem 8.8 (one dimension down) then connects them. We prove and mechanize the shared-face case (Theorem 8.12). For a general attachment, whose common boundary may span several shared faces, the same descent applies one transverse coordinate at a time, but we do not state a general theorem here: the cross-cell claims of this paper are the shared-face case, and the general zig-zag is left at the level of this sketch.

9. NO NATURAL STRICT FILLERS

De Morgan convexity coherently identifies same-cell fillers lying over one algebraic boundary. It is natural to ask for more: a coherent *choice* — a rule assigning to every boundary one of its strict fillers, stable under substitution. This section proves that no such rule exists, even without any coherence requirement between different boundaries.

9.1. Fiber combinatorics, recalled. We use the structure theory of boundary fibers developed in the companion paper [10], in its concrete representation: DM_n (the free De Morgan algebra on n generators, $\text{DM}(x_1, \dots, x_n)$ in the notation of Definition 2.1) is the lattice of monotone functions (up-sets) on the *literal cube* $L_n = \{0, 1\}^{2n}$, coordinates in pairs $(x_i, \neg x_i)$. A point of L_n is a *face point* if some pair lies in $\{(0, 1), (1, 0)\}$, and a *diagonal point* if all pairs lie in $\{(0, 0), (1, 1)\}$. The boundary ∂g of $g \in \text{DM}_n$ is the tuple of the $2n$ substitutions $x_i := 0, 1$. The relevant facts from [10]:

- (F1) fibers of ∂ are Boolean intervals $[b, b \cup D] \cong 2^D$ with b their least element and D an antichain of diagonal points;
- (F2) every fiber has a canonical least element $r(g) = g \setminus A_g$, where A_g is the antichain of minimal diagonal points of g (the *defect formula*);
- (F3) r commutes with a substitution f exactly when the transported boundary avoids the defect antichain — the *coherence dichotomy*; the minimal filler is thus not natural.
- (F4) (*sphere-filling*) every compatible boundary — a tuple of $2n$ elements of DM_{n-1} agreeing pairwise on their common restrictions — is the boundary of some element of DM_n : the boundary map is surjective onto compatible spheres. For self-containedness the statement is proved in Appendix A.

A substitution $f : [m] \rightarrow [n]$ of $\square_{\text{DM}}$ induces $f^* : \text{DM}_n \rightarrow \text{DM}_m$ and a map of literal cubes $F : L_m \rightarrow L_n$. A *section* is a function s from realizable boundaries to elements with $\partial(s(B)) = B$; it is *natural* if $s(\partial(f^*s(B))) = f^*s(B)$ for all f, B — i.e. the transport of a chosen filler is again chosen.

The representation of DM_n by up-sets goes back to the Berman-style description of free De Morgan algebras and is proved, in the form used here, in [10]. Everything below concerns DM_2 only, where (F1)–(F3) and every claim of this section are *finite* statements, each verified independently by the

census scripts of Section 12; the citation to [10] supplies the general structure theory, but the only general import anywhere in Sections 9–10 is the sphere-filling theorem (F4), used once, in the general clause of Theorem 10.2 — and for self-containedness we include its short proof in Appendix A, so that nothing unpublished is load-bearing.

The question left open by (F3): is the failure of naturality a defect of the *minimal* choice, repairable by a cleverer one?

9.2. The no-go theorem.

Theorem 9.1. *No natural section exists, already for $n \leq 2$ and already with respect to the endomorphisms of [2].*

Proof. Let $w = \neg x \wedge y \wedge \neg y \in \text{DM}_2$, the principal up-set generated by the point $\alpha = ((0,1), (1,1))$. Its boundary fiber is a singleton. Of the four diagonal points, $((1,1), (1,1))$ already lies in w , forced by upward closure from α , so it cannot be removed; a remaining diagonal point δ could be added only if every face point above δ already lay in w , and for each of the three one finds a face point above it missing from w (e.g. for $\delta = ((0,0), (1,1))$ the face point $((1,0), (1,1))$ is not in w), so none can be added without changing the boundary. So any natural section satisfies $s(\partial w) = w$, and hence is forced on every transport: $s(\partial(f^*w)) = f^*w$ for all f . Now

$$(x := y)^* w = y \wedge \neg y, \quad (x := x \wedge \neg x)^* w = (x \vee \neg x) \wedge y \wedge \neg y.$$

These two elements have the same boundary — their common fiber is the two-element interval with $D = \{((0,0), (1,1))\}$, of which $y \wedge \neg y$ is the top and $(x \vee \neg x) \wedge y \wedge \neg y$ the bottom — and they are distinct. Two different values are forced at one boundary; contradiction. $\square$

Remark 9.2. The proof is the coherence dichotomy (F3) weaponized: the diagonal substitution reaches the invisible diagonal point (its F maps a face point onto it), the connective collapse does not, and a *unique* filler upstream leaves no freedom to absorb the difference downstream. Machine census (Section 12): unique fillers force values on 42 of the 54 non-singleton fibers of DM_2 , with 142 pairwise conflicts of forced values; and independently of the forcing argument, an exhaustive backtracking search over all sections, constrained by 121 endo-substitutions of [2], finds no solution. The no-go is thus robust: it does not hinge on one clever element, and it needs neither higher cubes nor degeneracies.

To draw the categorical consequence precisely, fix what a stable filling operation is.

Definition 9.3 (Substitution-stable filling operation). A *filling operation* on a De Morgan cubical set X assigns to every boundary sphere $b : \partial \square^n \rightarrow X$ that admits an extension a chosen extension $\Phi(b) : \square^n \rightarrow X$. It is *substitution-stable* (natural in the cube category) if transported chosen fillers are again chosen: for every $f : \square^m \rightarrow \square^n$,

$$\Phi((\Phi(b) \circ f) \upharpoonright_{\partial \square^m}) = \Phi(b) \circ f.$$

This is the standing stability requirement on the filling operations of cubical type theories, where substituting into a filler must yield the filler of the substituted boundary. Note the condition concerns *full-boundary* extension — lifting against $\partial\Box^n \hookrightarrow \Box^n$, a trivial-fibration-style condition — and is not the open-box (Kan) filling condition; see Remark 9.5.

Corollary 9.4. *Over the single-edge computad — a generic edge $p : \text{Path } A a a'$ over distinct generic vertices — no substitution-stable filling operation on the cubes of the type theory can be valued in the strict layer. Equivalently: the representable $\Box_{\text{DM}}(-, [1])$ admits no substitution-natural choice of extensions for its extendable full boundaries.*

Proof. By the realization theorem the strict layer over the edge context is the full representable $\Box_{\text{DM}}(-, [1])$: strict n -cubes are classified bijectively by elements $g \in \text{DM}_n$ (the cube $A_p(g)$; constant cubes and vertex material correspond to the face-factoring elements), and their boundaries correspond to the boundary map ∂ of this section. Suppose every value of Φ were strict. Then $s(B) :=$ the classifying element of $\Phi(b_B)$ assigns to every realizable boundary B an element of its fiber, and the substitution-stability of Φ says precisely that the transport of a chosen filler is again chosen — i.e. that s is a natural section. Theorem 9.1 forbids this already over DM_2 and the endomorphisms of [2]. $\square$

Remark 9.5 (Scope). Two delimitations. First, the corollary is stated over the single-edge computad, where the realization theorem classifies *every* strict cube by an element of DM_n , so a strict-valued operation necessarily induces a section. Over a larger computad a filling operation might choose strict fillers originating from *other* cells for the fully degenerate boundaries (non-degenerate boundaries pin the cell by head separation), and extending the corollary would require an additional naturality of Φ under subcomputad inclusion; we do not pursue this. Second, full-boundary extension is stronger than open-box Kan filling: an open-box filler is not required to restrict on the missing face to the prescribed one, so an algebraic Kan structure does not obviously induce a natural section, and Theorem 9.1 is accordingly *not* claimed as a no-go for Kan fibrancy. What it rules out is any natural discipline for choosing among *existing* strict fillers.

Remark 9.6. The no-go sharpens the slogan of [10] — *without Eilenberg–Zilber normal forms, thinness cannot be a property* — into a theorem about *choices*: not only is there no property singling out a filler, there is no natural *choice function* either. It also gives the strict–weak divide studied throughout this paper and its companions [11] a purely categorical root: no natural boundary-extension discipline can live inside the reparametrization calculus. Whether this forces Kan *open-box* filling specifically — whether an algebraic open-box structure induces a natural full-boundary section — is an open question we record in Section 13.

10. THE MEDIAN CORRECTION STRUCTURE

If no filler can be chosen, all fillers must be carried — with the identifications between them. Section 8 supplied the identifications; here we show they are computed by a graded Boolean median and compose *on the nose* via iterated medians.

Proposition 10.1 (fiber coordinates). *Let $[b, b \cup D]$ be a fiber and $\varphi = b \cup S$, $\varphi' = b \cup S'$ with $S, S' \subseteq D$. Then the median interpolation (Lemma 8.3) in one fresh generator k is, as an up-set of L_{n+1} ,*

$$\begin{aligned} \chi(\varphi, \varphi') &= \text{cyl}(b \cup (S \cap S')) \\ &\cup \{(\alpha, \kappa) : \alpha \in S \setminus S', \kappa \in \neg k\} \cup \{(\alpha, \kappa) : \alpha \in S' \setminus S, \kappa \in k\}, \end{aligned}$$

i.e. the cylinder on the common part and the symmetric difference graded by the two half-cylinders of k : the Boolean median of S , S' and the k -grading, computed pointwise on D .

Proof. Evaluate $(\varphi \wedge \neg k) \vee (k \wedge \varphi') \vee (\varphi \wedge \varphi')$ at a point (α, κ) : for $\alpha \in b \cup (S \cap S')$ the third disjunct fires for all κ ; for $\alpha \in S \setminus S'$ only the first, i.e. $\kappa \in \neg k$; for $\alpha \in S' \setminus S$ only the second. (Machine-checked on all pairs of a rank-two fiber of DM_2 .) $\square$

Theorem 10.2 (coherence by iterated medians). *Let a, b, c lie in one fiber. Then the iterated median cylinder*

$$\Xi := \chi_l(\chi_k(a, b), \text{cyl}_k(c)) \in DM_{n+2}$$

has faces, on the nose:

$$\Xi|_{l=0} = \chi_k(a, b), \quad \Xi|_{l=1} = \text{cyl}_k(c), \quad \Xi|_{k=0} = \chi_l(a, c), \quad \Xi|_{k=1} = \chi_l(b, c).$$

More generally, every compatible sphere assembled from interpolations and cylinders over one fiber fills in DM_{n+j} : the correction system is strictly contractible, in the algebraically specified sense of Definition 8.7.

Proof. The four face identities are absorption computations from Proposition 10.1 (each face substitutes an endpoint into one grading; machine-checked exactly on a rank-two fiber). For the general statement, observe that a compatible sphere of corrections over a fiber has, at each original face of L_n , the constant value of the common boundary — so it is a compatible sphere in DM_{n+j} , and the sphere-filling theorem (F4) provides a filler, whose defect freedom is again a Boolean fiber. Contractibility at every level follows by induction, exactly as in Theorem 8.8. $\square$

Remark 10.3 (the pseudo-structure; an informal reading). The following is interpretation, not theorem; the precise statements are Theorem 9.1, Corollary 9.4, and Theorem 10.2. In the terminology of algebraic weak factorization systems [4], generated in $\widehat{\square_{DM}}$ by the boundary inclusions $\partial \square^n \hookrightarrow \square^n$: any section supplies one-step lifting data (the algebra-like data of the pointed endofunctor of the one-step construction), but Theorem 9.1

forbids the substitution-stability that a *monad*-algebra structure valued in strict fillers would require. The data that does exist naturally is: the full Boolean fiber 2^D of fillers (structure, not property), the defect antichain measuring every transport (F3), and the strictly contractible median system of Theorem 10.2 identifying all choices coherently. This is, informally, a fibered and strictly contractible pseudo-algebraic *analogy* — the coherent identification of all choices where no natural choice exists — and the no-go says nothing stronger is available. Steiner’s anticipation that thin fillers in weak settings “exist but not necessarily uniquely” [13] is thus realized in the sharpest form: existence, canonical non-uniqueness, and provable impossibility of natural selection.

11. THE STRICT–WEAK COMPARISON

The full theory adds the Kan operations. Two theorems close the comparison over a common boundary.

Theorem 11.1 (($\dagger$) is propositional). *Let $T := \langle j \rangle \text{hcomp} [j=0 \mapsto a, j=1 \mapsto b] (p j)$ be the constant-tube composite over a generic edge p — the left-hand side of the equation ($\dagger$) that the no-go theorem of [11] proves can never be definitional. Then the canonical filler*

$$\langle k \rangle \langle j \rangle \text{hcomp} [j=0 \mapsto a, j=1 \mapsto b, k=0 \mapsto p j] (p j) : \text{Path} (\text{Path } A a b) p T$$

is a path rel boundary from p to T : at $k = 0$ the added face is decided and the composite collapses to $p j$; at $k = 1$ the false face is discarded, leaving T . The same two computations give the statement for any constant-tube system in any dimension. (Machine witness `mixFILLD`, together with the standing definitional separation $T \neq p$.)

Remark 11.2. Theorem 11.1 is the exact dual of the transport-layer no-go theorem of [11]: *what can never be definitional is precisely what `hfill` supplies propositionally, uniformly.* The definitional/propositional boundary for composition is not a deficiency to be engineered away — the no-go theorems (there equational, here categorical: Corollary 9.4) say it cannot be — but a structural line, with the canonical filler standing exactly on it.

Proposition 11.3 (The weak layer extends the propositional classification). *In the circle instance, the composite `loop · loop` has winding number $+2$, while `refl`, `loop` and its reversal have winding $0, +1, -1$ (all machine-computed in the artifact library of [11]). Since winding is a propositional invariant, genuine composites are not propositionally equal to these strict cells: the weak layer adds new propositional classes over the same boundary.*

Table 1 now stands proved in all three rows: the first row, within one algebraic fiber, is Theorems 7.8 and 8.8, with the shared-face cross-cell extension in Theorem 8.12; the second is the no-go theorem of [11] together with Theorem 11.1; the third is Proposition 11.3. Combined with Corollary 9.4, the picture of the strict/weak boundary is complete from both sides — for pure

cells over a common boundary	definitional	propositional
strict, one algebraic fiber	unique up to strict homotopy	likewise
constant-tube composites	separated ((†) no-go)	strict, via the filler
genuine composites	separated	new classes (winding)

TABLE 1. The strict–weak comparison over a common boundary.

reparametrization and natural full-boundary extension (open-box filling remains the open question of Remark 9.5): *the strict layer of a cellular context is exactly its cellular presheaf, its canonical correction system is strictly contractible over each algebraic boundary fiber, and it cannot be enlarged — neither equationally (the (†) no-go of [11]) nor by any natural choice of full-boundary fillers (Theorem 9.1).*

For calibration, the machine-checked strictness catalogue of [11] sits exactly on this boundary: on the strict side, all connection reparametrizations, η -laws, and transport-related equations up to and including a definitional J on refl ; on the weak side, all hcomp -generated laws, beginning with $\text{trans refl refl} \neq \text{refl}$.

12. MACHINE VERIFICATION

All positive and negative instances quoted in this paper are elaboration-time checks in a self-contained cubical kernel written in Lean 4 (no dependence on Lean’s definitional equality), the artifact of [11]: repository

<https://github.com/r-shibusawa/cubical-strict-j>,

release v1.13.0 (commit c391d3d6), archived at DOI 10.5281/zenodo.21819833. A successful `lake build` replays every check; by (K3) each check is a decision, not a test. The probe files, by section:

- Sections 4–5: `LibSquares.lean` (the sphere and wedge probes), with the relevant rows of `LibStrictness.lean` and the proof notes `docs/ReparamCoherence2.md`;
- Sections 6–7: `LibGenBoundary.lean` (attachments, corners, edge squares, dependent unit and absorption laws, filler non-uniqueness) and `LibSharedEdge.lean` (cross-cell gluing and the no-merge separations), with `docs/GenericBoundary.md`;
- Section 8: `LibMixed.lean` (the interpolation homotopy with its control and separation, the machine-checked *failure* of the naive interpolant, the two deformation homotopies of the zig-zag, and the canonical filler of Theorem 11.1 with its definitional-separation control), with `docs/DeMorganConvexity.md` and `docs/MixedLayer.md`. The fiberwise-sharpness probes for Propositions 8.2 and 8.9 — `sqOrientCtrl` with its orientation-separation guard, and `sqTowerH`, `sqTowerH2` with the tower-separation guard — are in `LibSquares.lean`; the *impossibility* halves of the two

propositions are universally quantified over candidate tuples and are therefore pen-and-paper, resting on (K1)–(K2) alone;

- Sections 9–10: the census and search scripts accompanying the artifact. (1) *Forcing census*: over DM_2 , the unique fillers (singleton fibers) force values, via endo-substitutions from a generating set of 121, on 42 of the 54 non-singleton fibers, with 142 pairwise conflicts; the specific core of Theorem 9.1 is one of them, chosen for the shortness of its by-hand verification. (2) *Exhaustive search*: a backtracking search with constraint propagation over all sections — $2^{38} \cdot 4^{16}$ assignments for the 38 two-element and 16 four-element fibers — confirms unsatisfiability against the same 121 substitutions, independently of the forcing argument. (3) *Median identities*: the fiber-coordinate formula of Proposition 10.1 and all four face identities of Theorem 10.2 were verified exactly on a rank-two fiber of DM_2 . We remark that (1) alone proves Theorem 9.1, (2) alone re-proves it, and the printed proof needs neither.

The division of labor is the usual one for this artifact: the kernel supplies machine-checked *instances* (both identifications and separations); the theorems of this paper are pen-and-paper results about the conversion algorithm, whose load-bearing clauses (K1), (K2) are small, inspectable functions.

13. RELATED WORK AND OUTLOOK

Computads/polygraphs as presentations of free higher structures go back to Street and Burroni [1]; see [9] for a modern account. The present results interpret the *free* structure they present inside cubical type theory [3] literally: definitional equality itself realizes the presented cell complex over the De Morgan cube category [5, 2], with nothing added and nothing collapsed. The wedge and realization theorems are coherence results of “every diagram commutes iff it commutes in the free structure” type [8]: a diagram of reparametrizations commutes definitionally iff it commutes in the free De Morgan algebra — and, complementing it, no diagram involving composition commutes definitionally unless trivially. The uniqueness of Kan fillers up to homotopy is foundational for cubical models [3]; De Morgan convexity locates its exact strict shadow, running on the connection algebra alone. Winding numbers and the circle computation descend from the encode–decode method [6]. The weak ω -groupoid structure of identity types is [7, 14]; the present paper delimits, for the De Morgan cubical setting, exactly which part of that structure is definitional.

Three directions seem worthwhile. First, a *strict model structure in miniature*: De Morgan convexity gives the strict layer fiberwise contractible correction systems, and Section 10 identifies the pseudo-algebraic structure this supports; whether it organizes into a factorization system on cellular presheaves — with the strict layer as the “cofibrant skeleton” of the full theory — deserves study. The *quantitative geography of the obstruction* is part

of the same question: the forcing graph of Section 12 was here used only for unsatisfiability, but its structure should refine the coherence dichotomy (F3) into a cohomological invariant of the strict layer; the Kleene case [10], with its two polar fibers, is the natural test bed. Second, the *converse direction*: Theorem 9.1 shows a natural choice of full-boundary extensions requires values beyond reparametrizations; how far beyond? The first question is whether the no-go transfers to open-box (Kan) filling proper — whether an algebraic open-box structure induces a natural full-boundary section (Remark 9.5); if it does, the no-go becomes a statement about algebraic fibrancy itself. The minimal natural filling operations valued in Kan composites — and whether the CCHM `hcomp` is initial among them in a precise sense — is, we believe, the right formulation of the “value of the weak layer” question, and the strictly contractible median system of Section 10 is the comparison datum such a formulation needs. Third, *word problems*: since realization is exact, normal forms for the free cubical ω -category on a computad in the De Morgan signature can be read off from the type theory’s normal forms; the companion paper [11] carries this program out, locating the decidable–undecidable boundary of word problems over cellular contexts exactly at the strict–weak boundary studied here.

Declarations.

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Data availability. The kernel, probe suites and census scripts supporting all machine-checked claims are openly available in the repository named in Section 12, release `v1.13.0` (commit `c391d3d6`), archived at DOI 10.5281/zenodo.21819833.

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Use of AI tools. An AI assistant was used in the preparation of this manuscript; the author takes full responsibility for its content.

APPENDIX A. THE SPHERE-FILLING THEOREM

For self-containedness we state and prove the sphere-filling theorem (F4), following [10], in the up-set representation of Section 9: DM_n is the lattice of up-sets of the literal cube $L_n = \{0, 1\}^{2n}$, and for $i \leq n$, $e \in \{0, 1\}$ we write $L_n^{(i,e)} \subseteq L_n$ for the *face subcube* of points whose i -th pair is $(0, 1)$ (if $e = 0$) respectively $(1, 0)$ (if $e = 1$); it is canonically isomorphic to L_{n-1} , and the boundary map ∂ restricts an up-set to the tuple of its traces on the $2n$ face subcubes.

Definition A.1. A *sphere* in DM_n is a family $g = (g_i^e)_{i \leq n, e \in \{0,1\}}$ with $g_i^e \in DM_{n-1}$, satisfying the compatibility conditions $\partial_j^\delta g_i^\varepsilon = \partial_i^\varepsilon g_j^\delta$ for $i \neq j$ (indices adjusted as usual).

Theorem A.2 (Sphere filling). *Every sphere is the boundary of an element of DM_n ; a canonical filler is the up-closure m_g of the union of the prescribed points, which is also the least filler.*

Proof. Identify each g_i^e with its up-set inside the face subcube $L_n^{(i,e)} \subseteq L_n$, and let $m_g := \uparrow(\bigcup_{i,e} g_i^e)$, the up-closure in L_n . We must show $\partial_i^e m_g = g_i^e$. The inclusion $\supseteq$ is immediate. For $\subseteq$, let $\gamma \in L_n^{(i,e)}$ lie in m_g , say $\gamma \geq \delta$ with $\delta \in g_j^{e'}$. If $(j, e') = (i, e)$, monotonicity of g_i^e gives $\gamma \in g_i^e$. The case $j = i, e' \neq e$ cannot occur: the two prescribed literal pairs $(0, 1)$ and $(1, 0)$ are incomparable, so $\delta \leq \gamma$ fails. Hence $j \neq i$; let δ' agree with δ except that its i -th pair is set to the value prescribed by the face (i, e) ; from $\delta \leq \gamma$ and $\gamma \in L_n^{(i,e)}$ one checks $\delta \leq \delta' \leq \gamma$ componentwise, and δ' lies in both face subcubes (i, e) and (j, e') . Monotonicity gives $\delta' \in g_j^{e'}$, compatibility transports this to $\delta' \in g_i^e$ (both sides restrict to the same element of DM_{n-2} on the common subcube), and monotonicity again gives $\gamma \in g_i^e$. Minimality is clear: any filler's up-set must contain every prescribed point. $\square$

APPENDIX B. THE KERNEL INTERFACE: CODE CORRESPONDENCE

(K1) and (K2) are the complete interface between this paper and the kernel of [11]. For referees wishing to audit them against the artifact (Section 12) we record the correspondence; the relevant code is small and self-contained.

- *(K1), endpoint collapse.* Implemented by the path-application clause of the normalization-by-evaluation evaluator in the file `Cubical/Semantics.lean` of the library. Whether the interval argument is an endpoint is decided by the antichain-normal-form procedure of `Cubical/Interval.lean` — the free De Morgan algebra with reduced disjunctive normal forms of Definition 2.1 — not by syntactic comparison; on an endpoint the clause returns the stored endpoint (for `PathP`, the stored boundary cell), and otherwise it extends the neutral spine with the argument.
- *(K2), spine-wise conversion.* Implemented by the neutral-spine clause of the type-directed conversion checker in the same file: two neutral values are convertible iff their heads coincide, their spine lengths coincide, their stored endpoint annotations are convertible, and their interval arguments are equal in the free algebra (again decided by antichain normal forms).
- *Proof sketch for (K1)–(K2).* Both facts hold essentially by definition of the two clauses; the only mathematical content is the correctness of the antichain-normal-form decision — that reduced disjunctive normal forms are canonical for the free De Morgan algebra, so that the procedure identifies exactly the elements equal to 0 or 1 (for K1) and exactly the equal pairs of interval arguments (for K2). Canonicity of the reduced DNF is the classical normal-form theorem for

free De Morgan algebras [3]. What [11] adds is the verification that the two clauses interact correctly with the *rest* of the kernel — congruence with the other evaluator clauses and stability of the value representation — none of which is consumed by the arguments of this paper, which only ever apply the two clauses to spines of declared cell variables.

- *Audit path.* Each concrete instance cited in this paper is an elaboration-time guard in the probe files listed in Section 12; a successful `lake build` re-decides all of them against exactly these two clauses. The full verification of (K1)–(K2), and the proof of the termination and decidability theorem (K3), are in [11], which can be supplied to the referees; nothing beyond the two clauses above is consumed by the present paper.

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